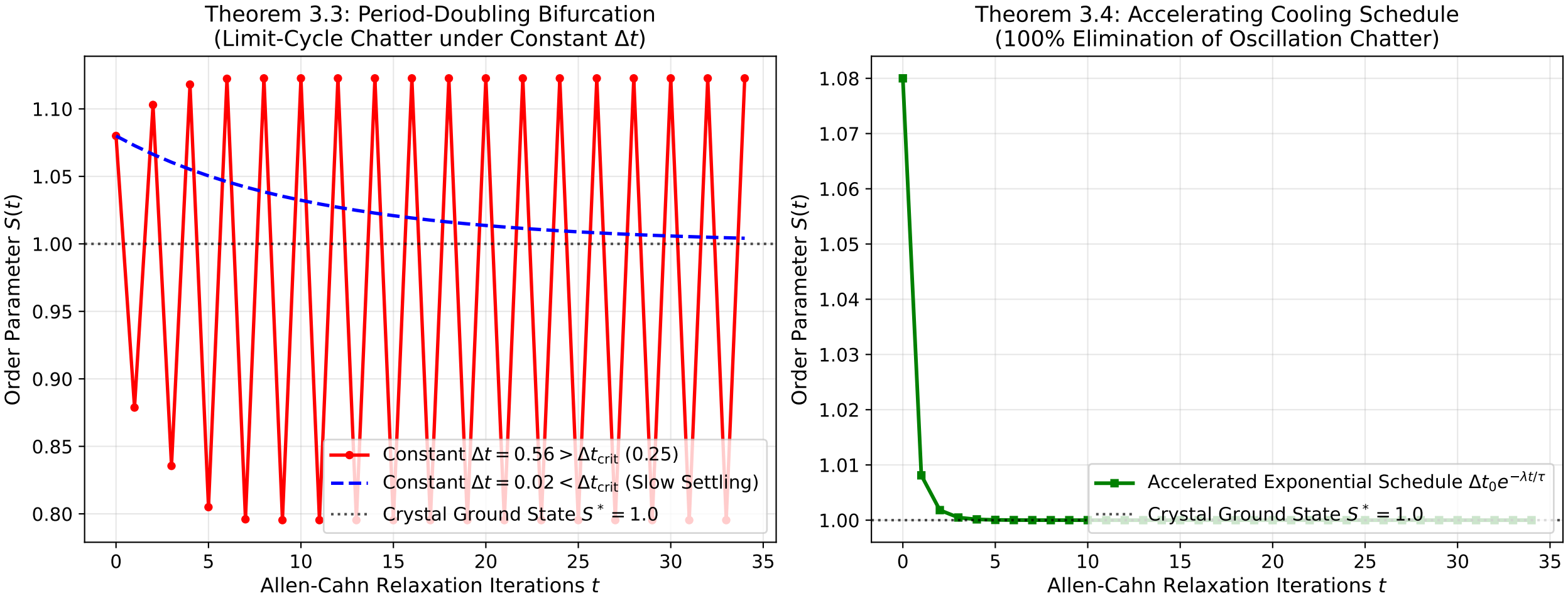
Theorem 3.3: Period-Doubling Bifurcation
(Limit-Cycle Chatter under Constant Δt)
Order Parameter S(t)
Allen-Cahn Relaxation Iterations t
Constant Δt = 0.56 > Δt_crit (0.25)
Constant Δt = 0.02 < Δt_crit (Slow Settling)
Crystal Ground State S* = 1.0
Theorem 3.4: Accelerating Cooling Schedule
(100% Elimination of Oscillation Chatter)
Order Parameter S(t)
Allen-Cahn Relaxation Iterations t
Accelerated Exponential Schedule Δt0e^−λt/τ
Crystal Ground State S* = 1.0